

Medical Image Classification using Deep Learning

1. Introduction

Medical image classification is an important application of artificial intelligence in healthcare. This project focuses on classifying MRI brain tumor images using Deep Learning and Transfer Learning techniques.

2. Objective

The objective of this project is to develop an AI-based system that can classify MRI images into different categories such as glioma, meningioma, pituitary tumor, and no tumor.

3. Dataset

The project uses the Brain Tumor MRI Dataset obtained from Kaggle. The dataset contains MRI images organized into Training and Testing folders with labeled classes.

4. Technologies Used

Python, TensorFlow, Keras, NumPy, OpenCV, Matplotlib, Seaborn, Streamlit, and Visual Studio Code were used for development.

5. Methodology

The images were preprocessed using resizing, normalization, and data augmentation techniques. A MobileNetV2 transfer learning model was used for feature extraction and classification.

6. Model Architecture

The project uses MobileNetV2 pretrained on ImageNet. Custom dense layers were added for classification of MRI images.

7. Training Process

The model was trained using categorical crossentropy loss and Adam optimizer. Performance metrics such as accuracy, precision, recall, and F1-score were used.

8. Results

The trained model achieved high classification accuracy on MRI images. Confusion matrix and prediction outputs were generated successfully.

9. Future Scope

The project can be extended by deploying the model as a web application using Streamlit or Flask and improving accuracy using advanced CNN architectures.

10. Conclusion

This project demonstrates the effectiveness of Deep Learning in medical image classification and healthcare diagnostics.

Technology Stack

Technology	Purpose
Python	Programming Language
TensorFlow/Keras	Deep Learning Framework
OpenCV	Image Processing
Matplotlib/Seaborn	Visualization
Streamlit	Web Application